

Ajay Kasturirangan

Jersey City, NJ | (929) 587-7900 | ajay.rangan@gmail.com
[linkedin.com/in/ajay-rangan](https://www.linkedin.com/in/ajay-rangan) | ajay1808.github.io | github.com/ajay1808

Summary

Data Scientist with 4+ years of experience building data pipelines, AI tools, and analytics dashboards. Proficient in Python and GCP to engineer data workflows. Builds models using quantitative methods, time-series forecasting, and anomaly detection. Develops AI applications utilizing RAG, agentic modeling, and LLMs to automate research. Visualizes findings using Tableau and Power BI.

Experience

Data Scientist / R&D Analyst, TIP Strategies

Aug 2023 – Present

- Built local RAG applications (Python, Ollama, LlamaIndex) to query 30 years of reports and summarize SEC filings, cutting research turnaround by 20%.
- Automated data pipelines via GCP and Python microservices and the Gemini API to generate dataset summaries and update client-facing Tableau dashboards, saving 3 hours per project and driving add-on revenue in 20% of engagements.
- Forecasted labor impacts using LSTM models and historical job data. Simulated expansion scenarios and reduced forecast variance by 40%, informing workforce planning for a \$17B regional investment.
- Built a geospatial transit framework (Python, NetworkX) to process GTFS data and score route capacity. Visualized networks in Tableau for 10+ projects, reducing analysis time from days to minutes.
- Deployed a local and custom OpenRouteService instance via Docker to process isochrone and drive-time queries, cutting processing time by 50% and enabling rapid site-selection analysis.
- Deployed Hugging Face zero-shot classification models to categorize grant data and build knowledge graphs, replacing a rule-based system and improving accuracy by 30%.
- Authored reusable Python API wrappers for public and licensed data sources including Census, BLS, BEA, and Lightcast, standardizing dataset ingestion across the firm's analyses, pipelines, and client-facing dashboards.
- Built a national freight data framework in Python to analyze supply chain networks and benchmark trade hubs, launching a new analytical product utilized in 3+ client projects.

Data Analyst, ICICI Lombard General Insurance Company

Feb 2021 – Aug 2022

- Integrated multi-terabyte datasets via Azure (Data Lake, Synapse) to migrate legacy SAP BO dashboards and develop 40+ Power BI financial reports, reducing report rendering time by 90% and centralizing operational tracking.
- Built topographic risk dashboards in Power BI using elevation and precipitation data to identify flood-prone infrastructure across 100+ miles of commercial highway projects, providing clients with realignment strategies to reduce insurance costs and support retention.

Selected Projects

- Data Center Siting Suitability Map** (GitHub)— Rebuilt a commercial data-center siting product as an open tool: scored 1,467,441 H3 resolution-7 cells across the continental US on 29 open federal and NGO data layers, and served the result as static PMTiles vector tiles in MapLibre GL with no backend.
- Site Selection Copilot** (GitHub)— Built a multi-agent system (LangGraph, Claude API) that answers "where should this business open next?" by routing to five geospatial and economic data tools - drive-time isochrones, census demographics, competitor density, wages, zoning - and fact-checks every figure in its output against upstream tool results. Picked the real chosen site in 7 of 8 validation cases.
- City-Scale ETA Prediction & Routing Engine** (GitHub)— Modeled a delivery network as a spatiotemporal graph on Uber's H3 hex grid rather than a flat regression over raw coordinates, with learned historical-edge-time priors, a distance-weighted KNN fallback for unseen routes, and a CatBoost model chosen for native categorical handling.
- Street Safety in NYC** (GitHub)— Rebuilt a 2022 crash-risk model after finding it trained on a sample 775x denser in crashes than reality; corrected the case-control sampling via a King & Zeng intercept shift, moved evaluation onto the full 7.5M-cell unsampled panel, and replaced accuracy with PR-AUC and lift@k against three baselines.

Education

New York University

Sep 2022 – Aug 2023

- M.S. Applied Urban Science and Informatics
- Graduated with a 3.9 GPA and received the Scholastic Award.
- Teaching Assistant:** Instructed students in supervised and unsupervised machine learning methods including ensemble methods and anomaly detection using Python, and supported a disaster management course utilizing Power BI for spatial and risk analysis.

B.Tech Electronics & Communication Engineering

- **Minor in Data Science:** Built a strong quantitative foundation through applied projects and coursework in statistical inference, network analysis, linear algebra, calculus, and R programming.

Skills

- **Programming & Libraries:** Python (Pandas, GeoPandas, Shapely, Rasterio, GDAL, NetworkX, Scikit-learn, TensorFlow, Statsmodels, PySpark), SQL, R (ggplot2, Leaflet, Plotly), DAX, Power Query M
- **Machine Learning & AI:** Generative AI (RAG, agentic modeling), Time-series (LSTM, ARIMA), NLP (Hugging Face), Predictive analytics, Anomaly detection, Ensemble methods, Clustering, Bayesian networks
- **LLM Stack & Frameworks:** LlamaIndex, LangChain, LangGraph, Ollama, Claude API, Gemini API, Vector DBs (ChromaDB)
- **Cloud & Data Architecture:** GCP (BigQuery, Vertex AI), Azure (Data Lake, Synapse), AWS (S3), REST API integration, Dask, Apache Spark
- **Visualization & Dashboards:** Tableau, Power BI, Streamlit, pydeck, Folium, Matplotlib / Seaborn
- **Developer Tools:** Git, Docker, LaTeX, Advanced Excel, STATA